INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **B.Tech** , Semester: **Second**

Stream: **CSE(IOT)/CSE(IOT, CS, BT)/ CSE(IOTCSBT)/CSE(Networks) /CSE /CSIT /IT /CSBS**

Paper Code: **BSCCH202**

Paper Name: **Chemistry**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. What is the significance of Ψ and Ψ^2 in quantum mechanical model of atom? **2**

2. Calculate the energy per photon for radiation of wavelength 500 nm. **2**

3. Which of the following atoms do not exhibit NMR spectra? **2**

C^{12}, H^2, N^{14}, N^{15}, O^{16}, C^{13}, F^{19}, P^{31}

4. Define the following term: Ligand. **2**

5. Which of the following compounds will exhibit optical isomerism: $CH_3CH_3CH(Br)CH_3(CH_3)_2$ **2**

6. Mark the asymmetric carbon atoms and give the number of optical isomers in the following compound: $CH_3(CHOH)_2COOH$ **2**

7. Define specific conductance. Mention its units. **2**

8. How is specific conductance related to equivalent conductivity and normally? **2**

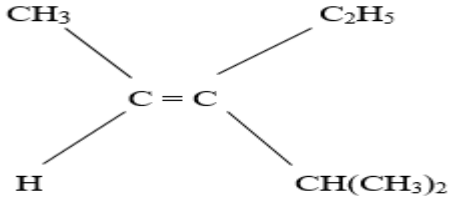
9. What is effect of dilution on pH value of HCl solution. **2**

10. Define transport number and ionic mobility. **2**

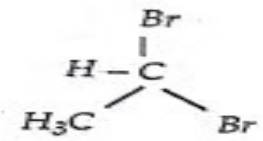
11. What is an electrochemical cell? **2**

12. What is a fuel cell? **2**

13. Assign E or Z configurations to the following component: **2**



14. Assign R and S configuration of the following compound: **2**



15. What is the shape of methyl carbocation? **2**

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Why is Heisenberg's uncertainty principle ignored in everyday life? **5**

17. What is meant by electromagnetic radiation? Describe some important characteristics of electromagnetic radiation. How do the wavelengths and frequencies of different type of electromagnetic radiations vary? **5**

18. The microwave spectra of CO molecule gave $\frac{2h^2}{8\pi^2 I}$ =7.6 x 10^{-16} ergs. Calculate the bond length. **5**

19. One litre of a gas of 200 atm and 473°K is compressed to a pressure of 400 atmosphere and 273°K. The compressibility factors found from the curves are 1.072 and 1.375 respectively at the initial and final states. Calculate the final volume. **5**

20. Discuss in brief Bronsted-Lowry concept of acids and basis with examples. What do you understand by conjugate acid and conjugate base. **5**

21. Why is salt bridge used in the construction of a cell? **5**

22. Write the possible optical isomers in tartaric acid. **5**

23. Which among propan-1-ol, propan-2-ol, 2-methylpropan-1-ol, 2-methylpropan-2-ol reacts fastest with HBr? **5**

24. The treatment of alkyl chloride with aqueous KOH leads to the formation of alcohols; whereas in the presence of alc. KOH, alkenes are formed as the major products. Explain. **5**

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. What are first, second and third ionisation energies of an element? What are the factors on which ionisation energy of an element depends? **10**

26. Deduce expression for kinetic energy of a particle in 1-dimensional box. **10**

27. Why do the wavelength and extinction coefficient increase with an increase in conjugation in a compound? What sort of solvents are generally used in *UV* spectroscopy and why? An increase in solvent polarity shifts $\pi - \pi^*$ band to longer wavelength but $n \rightarrow \pi^*$ and $n \rightarrow \sigma^*$ bands to shorter wavelength – explain. **10**

28. Calculate the pH of : **10**
a) 0.0001 M HCl solution.
b) 0.04 M HNO_3 solution, assuming complete dissociation in each case.

29. 10 gms of CO_2 are contained in a 10 litre globe at 40°C. Calculate its pressure **10**
a) as a van der Waals' gas,
b) as an ideal gas. (a = 3.59, b = 0.043)

30. Describe *two* methods used for resolving racemic mixtures into optically active compounds. **10**

31. Will N-methylacetamide undergo Hofmann rearrangement? Explain. **10**

32. By using appropriate examples, discuss and illustrate the stereochemical implication of S_N1 and S_N2 reactions. **10**

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